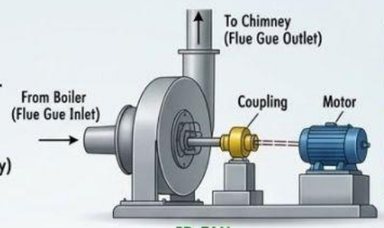
(BOILER ID FAN, FD FAN, PA FAN, SA FAN BASIC DETAIL)

① ID FAN (INDUCED DRAFT FAN)

ID fan is located at the outiler of boiler and creates a negative pressure in furnace. It suckes the flue guees from boiler and discharges to chimney.

- Function : To draw flue guees.
- Location : After boiler (Before Chimney)
- Pressure : Negative Pressure
- Duty : Gas handling
- Construction : Centrifugal / Axial
- Materials : –



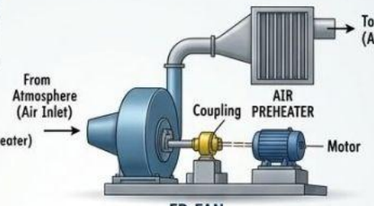
ID FAN

FAN COLOR KEY
FULL NAME
ID FAN
FD FAN
PA FAN
SA FAN

② FD FAN (FORCED DRAFT FAN)

FD fan supplies the required quantity of air to the furnace for proper combustion. It forces the air through air preheater and supplies to beller.

- Function : To supply air.
- Location : Before Boiler (After Air Preheater)
- Pressure : Positive Positive
- Duty : Air handling
- Construction : Centrifugal / Axial
- Materials : MS, CI



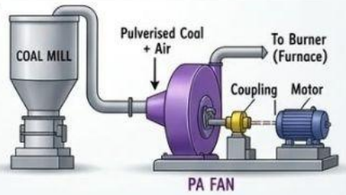
FD FAN

Note :- FD fan forces fresh air to furnace for proper combustion.

③ PA FAN (PULVERISED COAL AIR FAN)

PA fan (calso called Coal Mill fan) suplies primary air with coal to the burners. It carries the pulverized coal from mill to furnace.

- Function : To transport pulverized coal with air.
- Location : From Coal Mill to Burner
- Pressure : Positive Pressure
- Duty : Air + Coal handling
- Construction : Centrifugal
- Materials : MS, CI (Wear resistant)



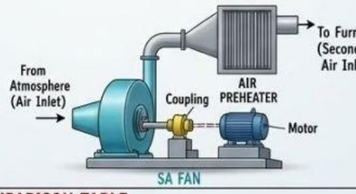
PA FAN

Note :- PA fan carries pulverized coal with air and delivers to burner.

④ SA FAN (SECONDARY AIR FAN)

SA fan supplies secondary air to the furnace above the fire zone for complete combuition of gazes.

- Function : To supply secondary air.
- Location : After FD fan / Air preheater
- Pressure : Positive Positive
- Duty : Air handling
- Construction : Centrifugal / Axial
- Materials : MS, CI



SA FAN

Note :- SA fan applies air above five zone for complete combustion.

COMPARISON TABLE

FAN	FULL FORM	FUNCTION	LOCATION	PRESSURE
ID FAN	INDUCED DRAFT FAN	Drows Flue Guees	After Boiler (Before Chimney)	Negative
FD FAN	FORCED DRAFT FAN	Supplies Air to Furnace	Before Boiler (After Air Preheater)	Positive
PA FAN	PULVERISED COAL AIR FAN	Transports Coal + Air	Coal Mill to Burner	Positive



FORCED / INDUCED DRAFT (FD/ID) FANS MAINTENANCE FRAGNET

Routine/Weekly Checks

- Lubrication: Verify grease/lube oil levels, operating temperatures, and flow to bearings.
- Vibration & Noise: Monitor for unusual noises or vibrations
- Temperature: Check bearing and motor winding temperatures.
- Leaks: Visually inspect for air, oil, or cooling water leaks.

Monthly/Detailed Checks

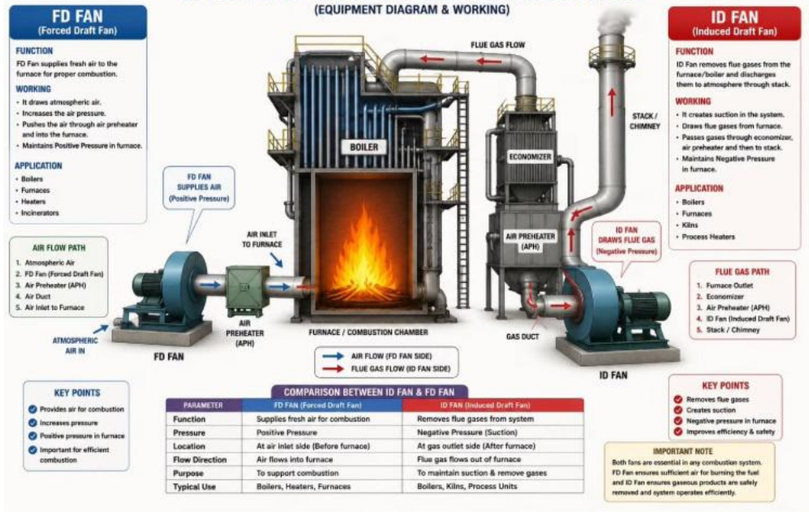
- Impeller & Blades: Clean accumulated dust and inspect for pitting, erosion, or corrosion.
- Belts & Couplings: Inspect V-belts for wear and correct tension. Check coupling alignment and bolt tightness.
- Fasteners & Mountings: Tighten foundation bolts and inspect static parts (ducts, inlet cones).

TA/Shutdown Overhaul

- Clearance Check: Verify the clearance between the inlet cone and the impeller using inside calipers.
- Shaft & Bearings: Inspect shaft seals. Measure bearing clearances and replace if worn.
- Coupling/Motor: Decouple the motor to check base bolt locking and perform alignment checks.
- Dampers: Check the open/close operation of suction/discharge dampers and Inlet Guide Vanes (IGV).
- Body corrossion check and refurbishment
- Bonnets painting or replacement
- Shafts alignment checks
- Bearing and couplings change
- Internal checks on dampers and cleaningg of deposited soot and other choking materials
- Belts replacement
- Motors function checks or rewinding
- Cabling and connection continuity checks

ID FAN & FD FAN IN CHEMICAL INDUSTRIES

(EQUIPMENT DIAGRAM & WORKING)



COMPARISON BETWEEN ID FAN & FD FAN

PARAMETER	FD FAN (Forced Draft Fan)	ID FAN (Induced Draft Fan)
Function	Supplies fresh air for combustion	Removes flue gases from system
Pressure	Positive Pressure	Negative Pressure (Suction)
Location	At air inlet side (Before furnace)	At gas outlet side (After furnace)
Flow Direction	Air flows into furnace	Flue gas flows out of furnace
Purpose	To support combustion	To maintain suction & remove gases
Typical Use	Boilers, Heaters, Furnaces	Boilers, Kilns, Process Units

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Status of Approval :- 01			
Dimentions as Mentioned NA	Drw electronic code	Drawn By: DFT-01	Drawn Date 20260731
Drawing Scale :- NTS	rPlan AT 39, MERMAID 1 CHS LTD., SEC-11, CBD-BELAPUR, NAVI MUMBAI - 400614	Checked Approved by rPlan-01	Checked Approved Date 20260810
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Angle of Projection NA / Sheet Size A3			